

A Packing of 12 Four-Spaces in $\mathbb{R}^{13}$

Limin Ge*

October 9, 2026

Abstract

We give 12 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.020671. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.999702, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 12 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.020671 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 12$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.999702, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 12/11 = 3.020979$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N12.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
188791968913951437013036806888587509256175418009974394756178282324223131
695708840619777615422496195128798838329131574132511959765163938580618066
971419510184494825000212147145114367773490356728
```

denominator:

```
624999999998392433004721853440215010282970067801034977892460318201613488
442468091110591506627997422496496134952760081349969020509791361662536933
64001329908999435416028280133560733546595029525
```

B Coordinates

12 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N12.txt`.

```
-93505574485 -111262460054 -343695165426 1328376333456139656785 -126914270703 -129440085528 -19975566309293758352 131733199921 506483897361 -53787048838
-226350988885 -88009316830 70872923924 -1408116795006588424820 -120306630942 -209084306922 -3385322656055182791806 -228980789051 319906271780 402734445478
242299044195 -320086182702 232667039900 -162019765200904614317 128030630940 462506695938 -31753953674540010123052 155601475535 -314328604436 -326150541815
105242642781 -5740890401 49647127320 -30435217035 122000090266 -299877254493 517406862876 2932180104689793741786 -70636468997 -83339274872 452346020664
-537023254750 43827983862 -27838909012 -1044953192561727456748 -507642569493 118680807010 -610887588465699973891 207189369683 -281073479824 -206715577920
-540069180043 311453958527 7860655181 -231974407478173432135561 284829546150 -129735421416 -48871045822993683637 89828752830 -232249770453 124672012320
128109582486 -232304271982 -122151305128 -295931811024618061982 381429038241 -9205834196 497261430326-224088648549 539913311806 75083850666 190385891360
151061838959 313886144907 483879735192 -32285798573632773291 474794855821 505218266325 122221133803 165350624001 -354094200618 151573016751 -283691073655
-353514240737 -580358377546 425122784935 249499975681012561178 226478666267 -238250250480 15100257469209310747306 443462041762 -79487636905 -26430767802
95801766305 47553670531 -544518425877 -38721871986 215414903030 152259284928 31431084979 -185802389620-46421650379 -368422428869 -579189025929 131854630105
-50870913581 -434049619928 -163911275164 -748371645636640640570 -185370129364 78061417637 189998958122527577048 257124077605 -163427389240 371563757658
223905365396 214677790079 255028536611 -283761153378445471811303 -163222596654 275704904182 -221707406812430718897 43021264395 71702391614 -437803194679
-253110445599 228931936775 34905300151 -2688605865206545286382 -133695849352 -178331656512 3532544509601533739099 181024352974 14281282147 -33218601890
71130534604 221033385367 357965002145 -372194939305 183336361155 -274671303693 -185565712942 97936734422245176814 37402864407 287951398327 5678835839
134860947868 -436997538590 82950144298 -40004421427383887068646 -305084176941 499878138450 328670753733684013468 -110560641586 150510855709 -61944247017
-254380243769 228835868094 -567573457583 -1646578746394735355437 54509603970 -137514590385 56599692293517597357309 226694440672 533038528869 -102536762453
20038073019 -385021736223 2047770620 99855192456 21157039447 -293203440364 147574313156 -4796925723622309675806 -133718555531 85425284780 133465128188
1716190904862 -495234988232 -236592713925 -41669313231618173117 -409307706537 100152603039 241583825321884456982 326636299426 251529057326 -348232521949
-258862985286 45197866690 -276351415670 -109451755796242696866 -37235504468 281191060996 -2490861848405557968877 -60976310807 130167567696 -31944235956
258377661710 -70441608846 -226846918025 46076605239823708387940 -98795742503 -140738005943 -10995902350177032228482 243436345594 -469934400942 -361645553951
374348912050 371068812803 123598609600 -2133595276935843215732 -540263353881 -31131821759 82082418025 270492405819 -242046868738 -164009354552 -155678116877
210401226335 163890898806 -492355050952 -44222241908218433851164 -482453826649 -190753291667 -27639127434701760380 -45865184488 71924600606 630712263421
562576242126 92088438243 -24472655358 82550989240 -145742982894 -29772495764 -265548459694 1520992926706273970300 -644810783693 108849228238 -334001976898
-379749794896 -116041933520 274233603366 -26086227305523913597 -137264681627 -625017930184 54663288009116720838 -391489955540 19162674891 -276562113774
301207064197 -21902881623 114802671502 -14119532139625505641298 49791247577 -259941173633 -82884476382598746206025 -128783250584 490846783438 -30215589640
104863856395 -346881598396 -111166632982 -320075051336648687997 -130409111357 -1561808378 325898571056-333429010431 -317985773743 -129367594352 322083346000
-45175379062 -28064068853 -47951634279 28583667045 122477668812 176395988685 -443656010790 -326864604130735900231 -480404761169 -64091052361 185895267753
678130097170 428409994245 119767994554 59591768258 -184658729369 -38174588794 466672703013 -79149796088124238609296 -6386337580 37902575351 -1715059348267
-3826798825 17450226187 -287785313623 -18090362230650700064175 245028918019 28169827791 -17663263666 -295991972681 10405867123 -181270436456 661980933676
323025438437 -143778167119 -395808016977 -395234007832020670151 583836360239 -254983106874 3604092035598903438089 -142803232380 -225637469564 -249035290045
-387172356611 332860932845 -331647803932 3059167667222814300501 214790787517 66986980739 8985206750 440007983758 86321081963 97871605535 177645108388
85021265187 -338415447070 318870620129 62282766905749287666504 -299892726064 364058155963 35975883178354633339571 -598652046890 286273607909 43625532921
78211743610 -98435508769 -454788140726 473530966729-259031870612 228029466609 271341314377 8936823767761247474296 -291742454216 -304918339848 357201839879
23994616528 669956325099 -22969831106 191148125129 88808365359 -85400565908 -69974838222 -94073664173 -56126297677 -46444239290 483712208174 85053881263
114361691261 -67698256469 153503279190 -47729790995-77221856329 402013936414 223754071001 90040075142 156198970176 382122383978 135484056389 29182944294
-129271683051 -78742419416 -34833172216 -173807416787574753048 352461042192 424727262187 16097568986 55192161947 -118988840980 629405919487 -33261120067
-15090064502 140635721206 509796006805 -77612541932-61049540209 122571863651 -61357706107 136335324848-93008276918 294165228204 74283542553 241527943464
-385891218143 280565766298 72729760208 -17678123554288323314590 -56077467747 -254278031137 6702675906994905469367 -161860812729 -265523946015 -453956891374
-301637702717 -39613039978 175274326290 -1577045723838584080724 -259131492517 -71739864972 3662404100931240166141 -150516247300 40885416649 -33652415441
-291541725849 195246093683 -97397175985 55457171062423341537655 112504775667 442821521773 -18258581983378521631233 53815335597 -285042932825 -103195807937
-250172424391 -139440769868 -4391046947 2909635175372254881262 -398445455332 -306440757748 248939391655320352137 -51948712059 -16448571044 -101114519594
2300400317 -123501732781 138463265151 -11767015062201676740333 -147385953342 63552185960 -96094893322-8189972534 274843636184 -307665401251 -95840245449
-100704057442 562260236118 49230803735 -2145951325317120447835 70271539773 205610043137 -249883170301-488364165101 623351303194 -42859342500 -245995543285
494579031161 225885043597 -189886819518 3746261168167270424816 252373563110 45182973755 393528563640 -391770276157 -578573004927 220722794060 -55242665045
69370288521 527613810144 514137906041 77374780409 -46373124167 -73503466037 208950553766 -84811346675-19803334539 266343162333 157809301458 300813254323
-249828212176 279335924790 -76983268966 340935893256208685809958 -627581336788 -155225867715 -2190517330282843921 -114070756128 -266807629467 -415188983336
75452764021 268093083000 -440617014312 -210913607942015618625503 -81275185827 -338434973223 -52268469537929188208 37359382653 -168284143053 362085357712
-240159306503 43734026167 -189406104639 -2194960742826874448452 24401006160 -217842121716 25264554405866342763827 208631511916 525852190366 28091654769
-288686551470 -237352161298 -93800656496 1728380560935044795130 23359336519 249226933388 73291341166 70664381054 -28440336155 120028828853 -821172336299
189932190030 -196994281957 604429183436 8817221054426722459608 313551525116 -511219036163 -351268228326730150630 -18080462794 -22301317708 -45885391805
-171451142850 185622298349 112659555483 14224890306970469229997 -353735980627 99216411957 19653039565709694662381 -77561781159 388877301088 -18963349960
-564142289012 15232845699 213855428884 -52318291955260876213 -328297484992 313999366027 -40525545524148120543679 -243954772658 -454099666659 -74020972158
```